

Transformers

Carlos Cotrini

CAS Building ML/AI Applications, ETH Zurich, weekend 2

Friday 18 September 2026

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Section 1

Multi-headed attention

Multi-headed attention

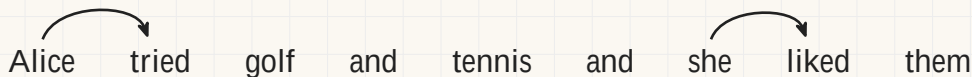
Remember that we need multiple attention maps.

Alice tried golf and tennis and she liked them

Multi-headed attention

Remember that we need multiple attention maps.

Subject-verb
relation



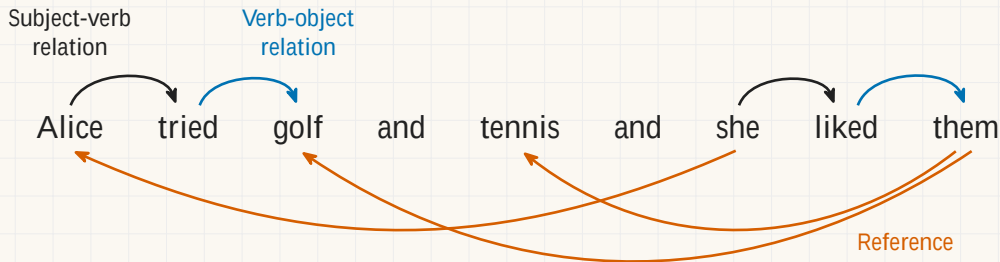
Multi-headed attention

Remember that we need multiple attention maps.



Multi-headed attention

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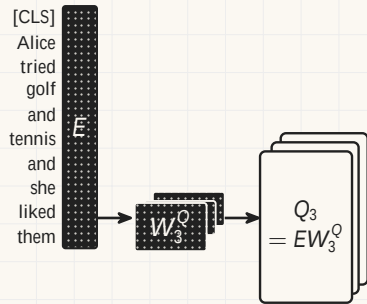


Multi-headed attention mechanism

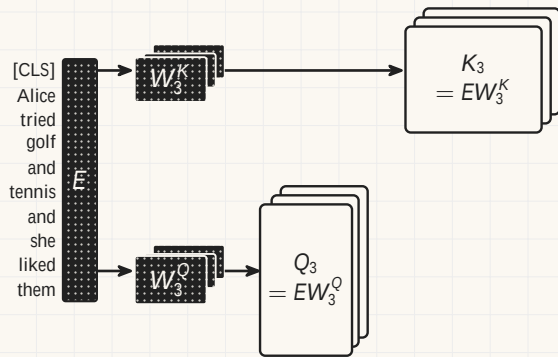
[CLS]
Alice
tried
golf
and
tennis
and
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liked
them



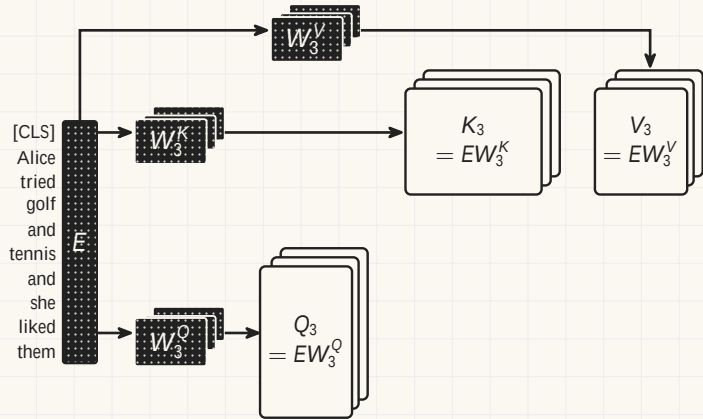
Multi-headed attention mechanism



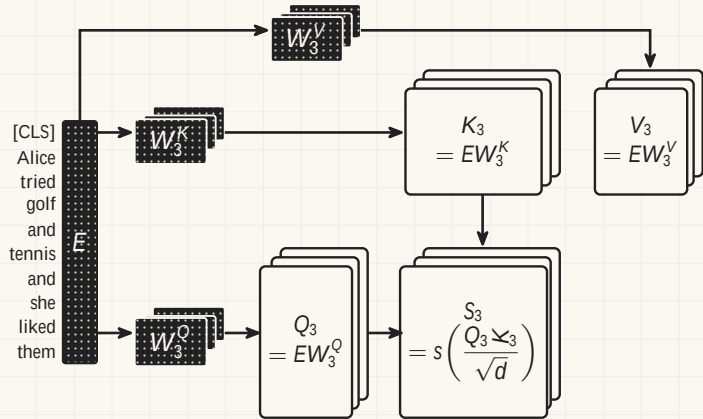
Multi-headed attention mechanism



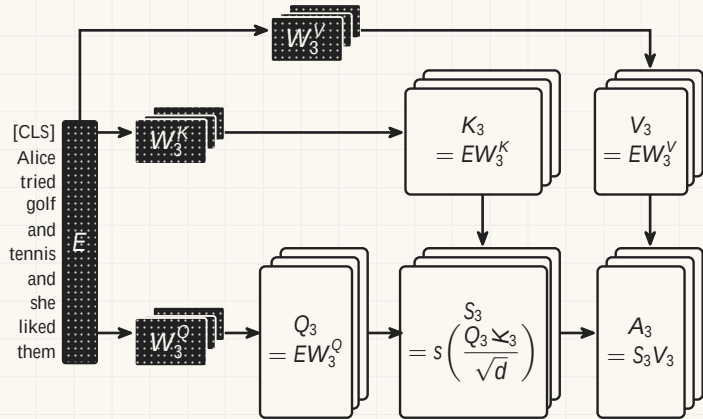
Multi-headed attention mechanism



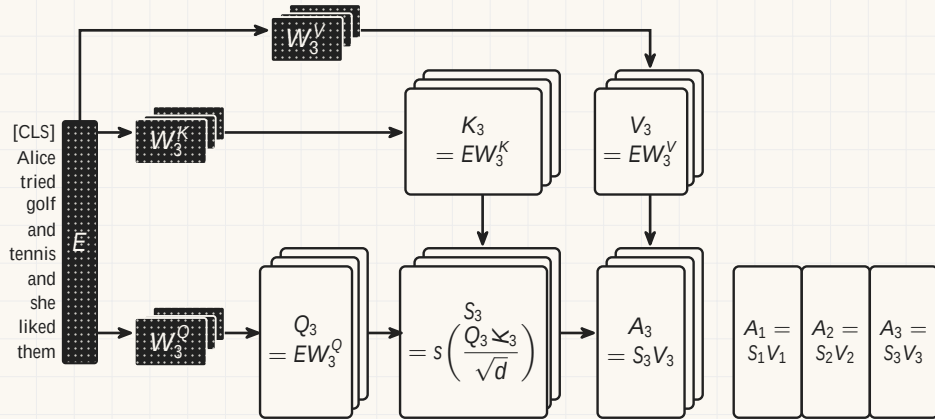
Multi-headed attention mechanism



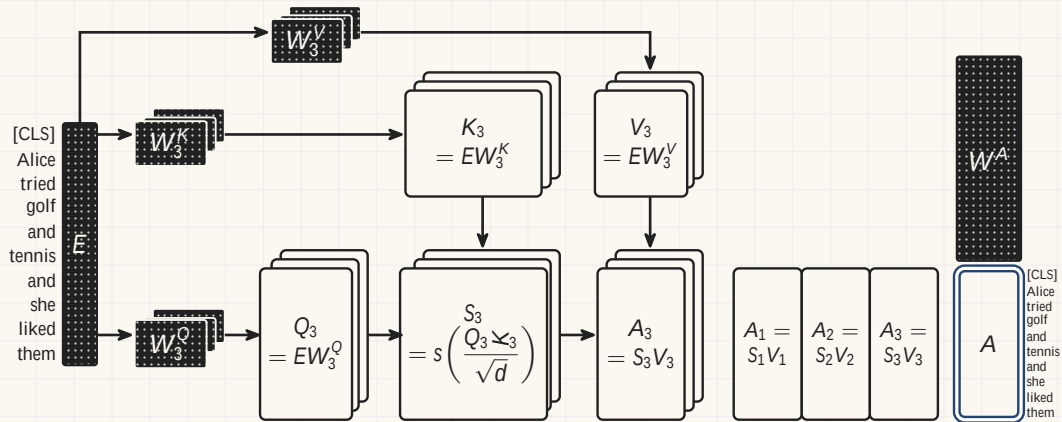
Multi-headed attention mechanism



Multi-headed attention mechanism



Multi-headed attention mechanism



2

Section 2

Cross-attention

Cross-attention

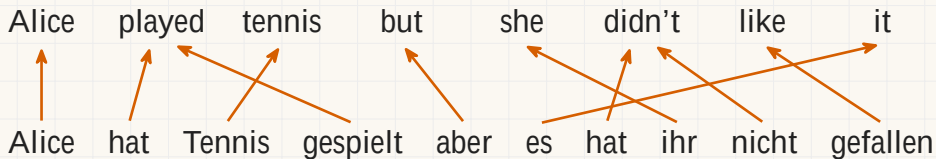
- Consider the problem of translating from English to German.

Alice played tennis but she didn't like it

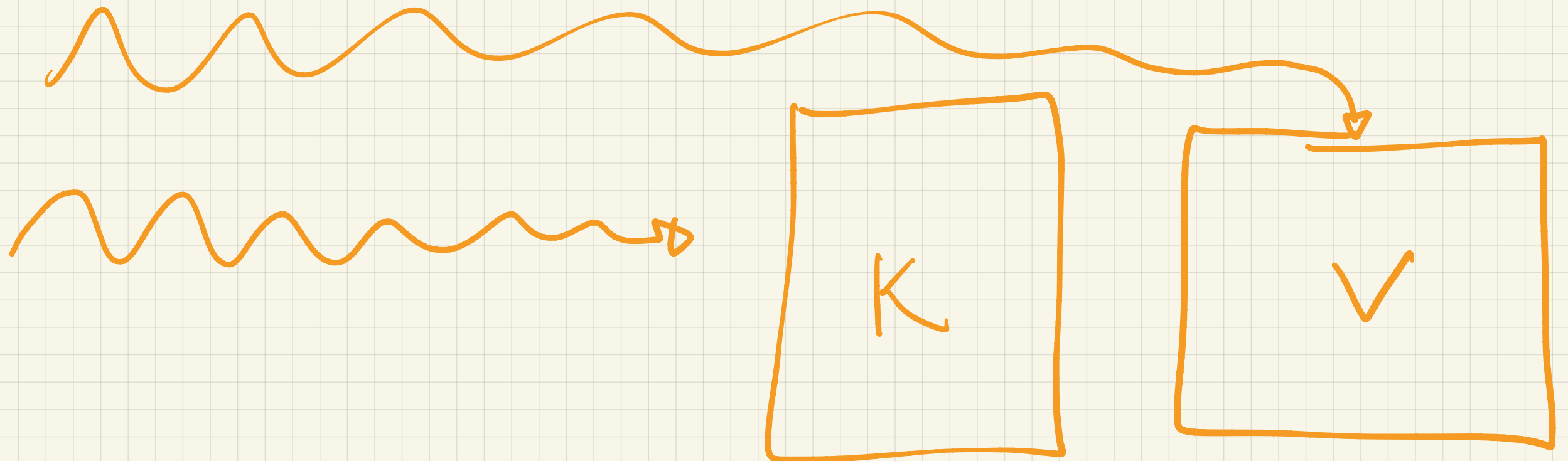
Alice hat Tennis gespielt aber es hat ihr nicht gefallen

Cross-attention

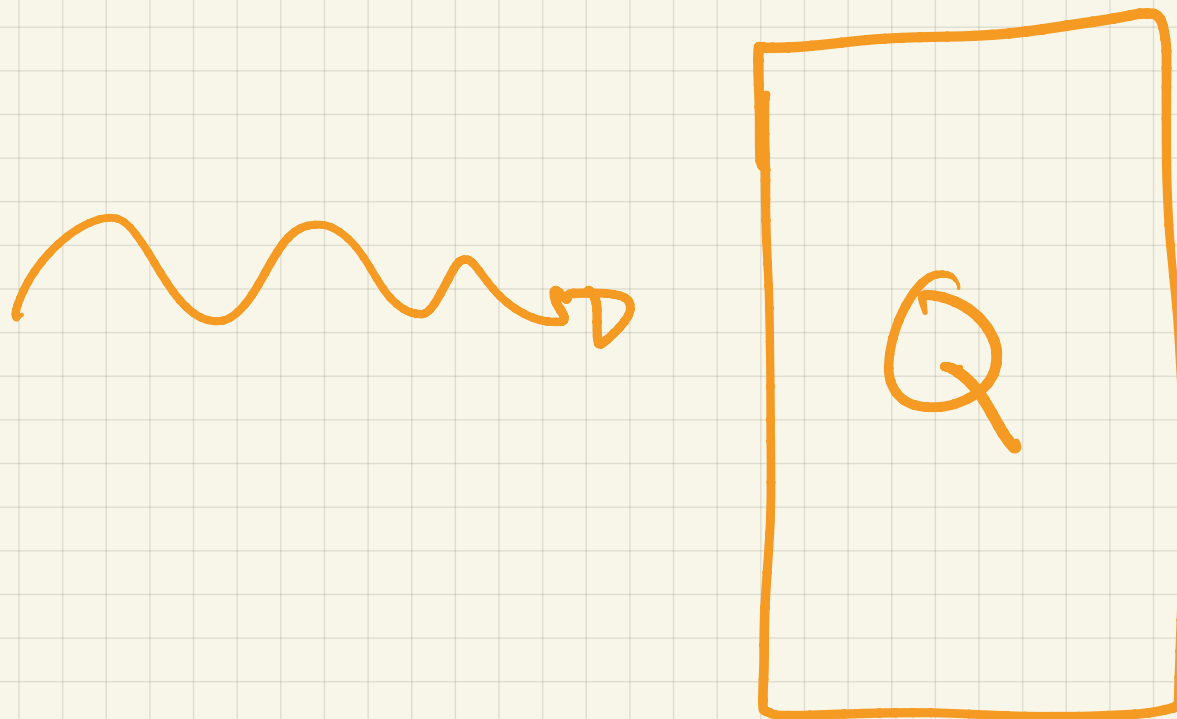
- Consider the problem of translating from English to German.
- A different attention mechanism is needed.



Alice
played
Tennis
but
⋮



Alice
hat
Tennis
gespielt
⋮



Abstraction of a cross-attention mechanism

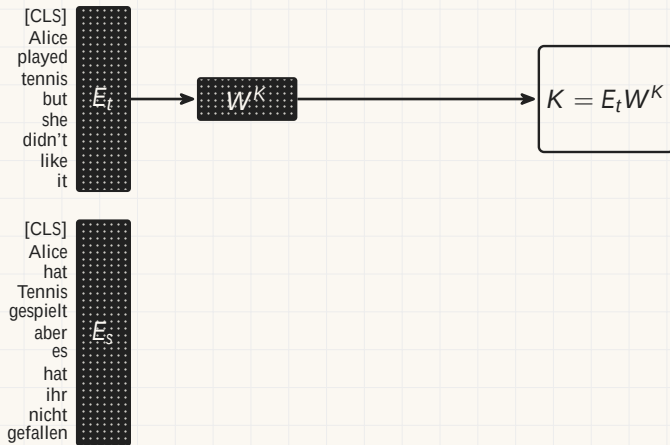
[CLS]
Alice
played
tennis
but
she
didn't
like
it

E_t

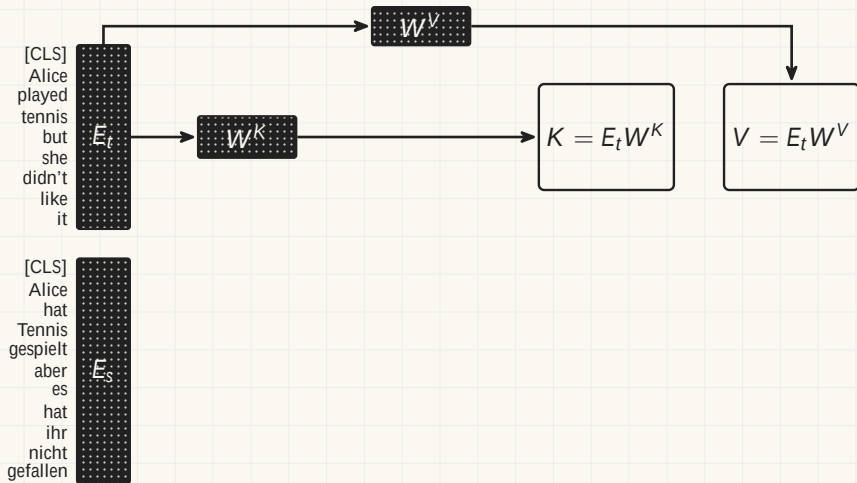
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Alice
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E_s

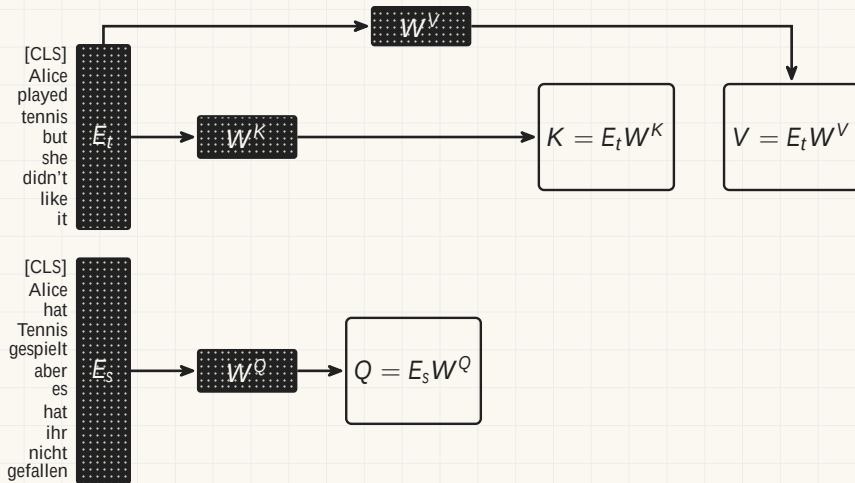
Abstraction of a cross-attention mechanism



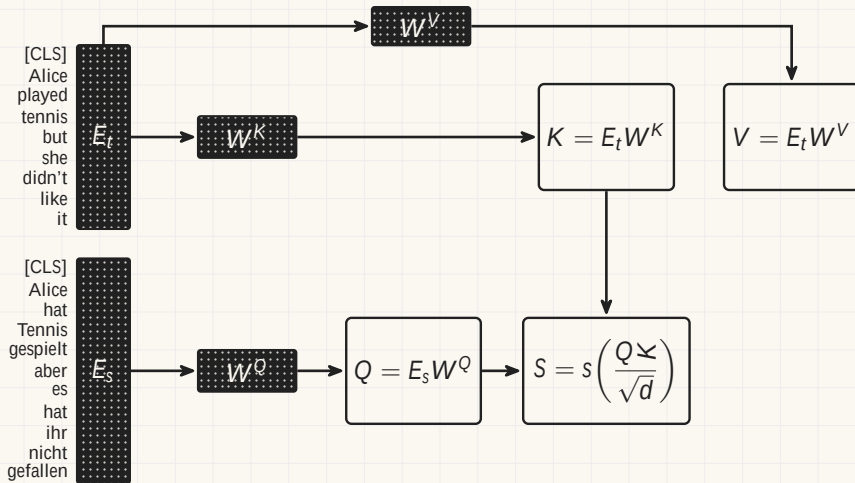
Abstraction of a cross-attention mechanism



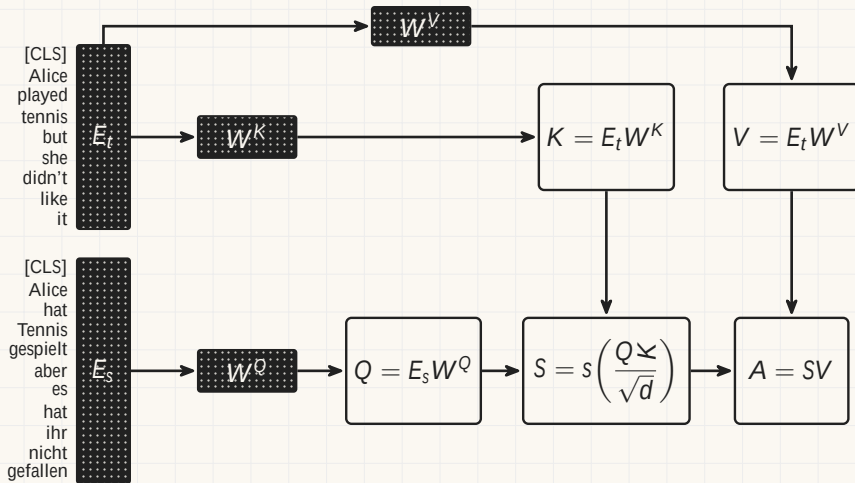
Abstraction of a cross-attention mechanism



Abstraction of a cross-attention mechanism



Abstraction of a cross-attention mechanism



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Section 3

Masked self-attention

Masked self-attention

- Consider the task of generating text for answering questions.

Why did the bicycle fall over

Masked self-attention

- Consider the task of generating text for answering questions.

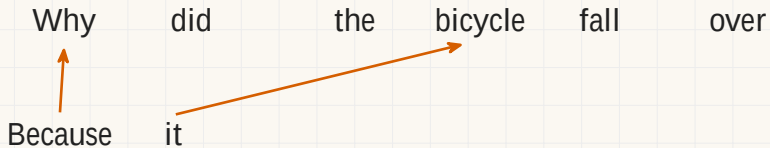
Why did the bicycle fall over

↑

Because

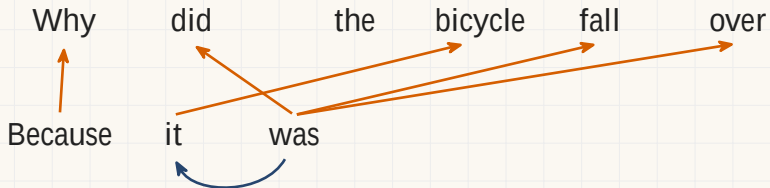
Masked self-attention

- Consider the task of generating text for answering questions.



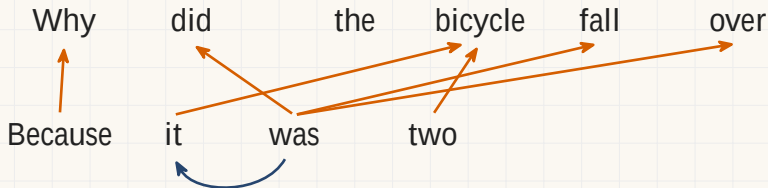
Masked self-attention

- Consider the task of generating text for answering questions.



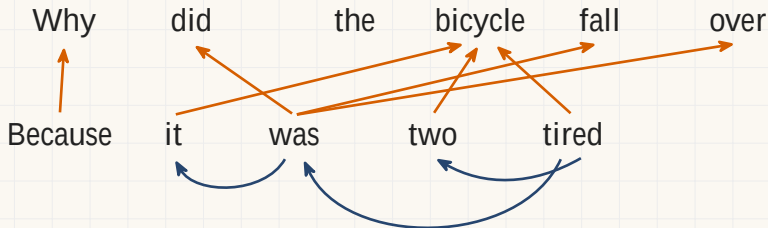
Masked self-attention

- Consider the task of generating text for answering questions.



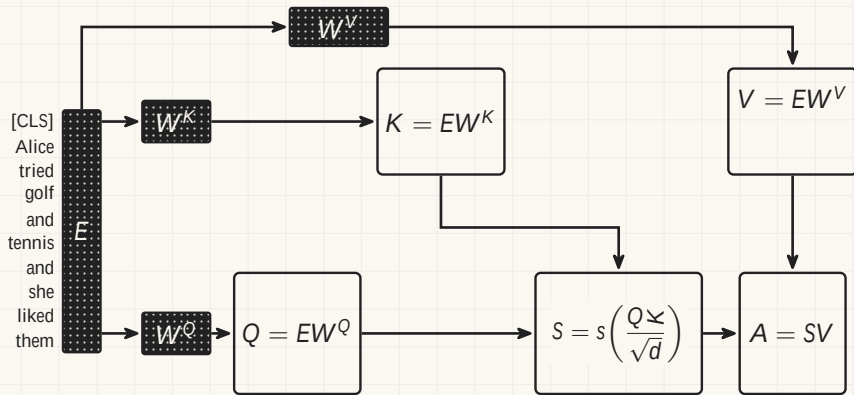
Masked self-attention

- Consider the task of generating text for answering questions.

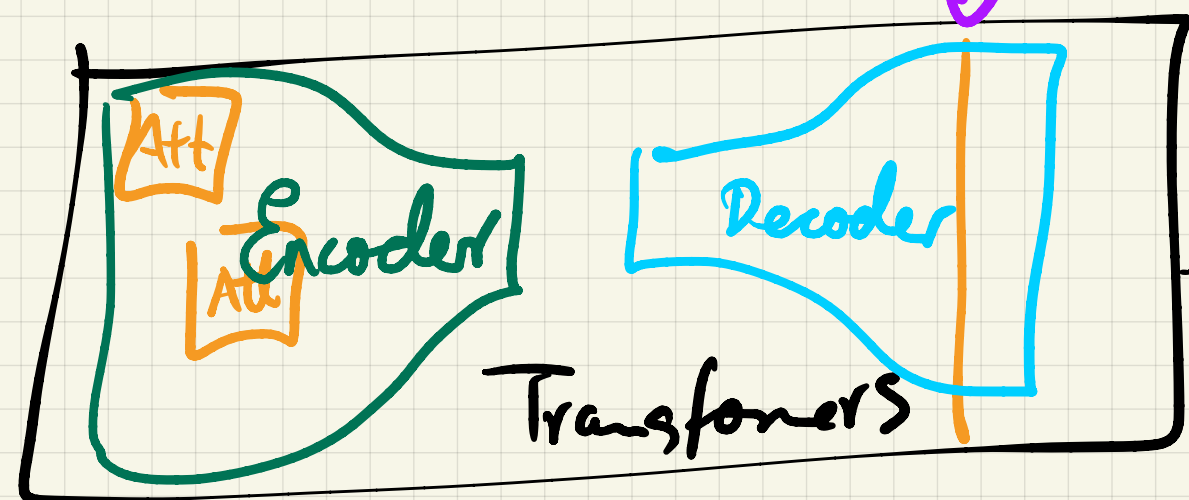


For generating sentences, we need an attention mechanism that only looks at what has been produced so far.

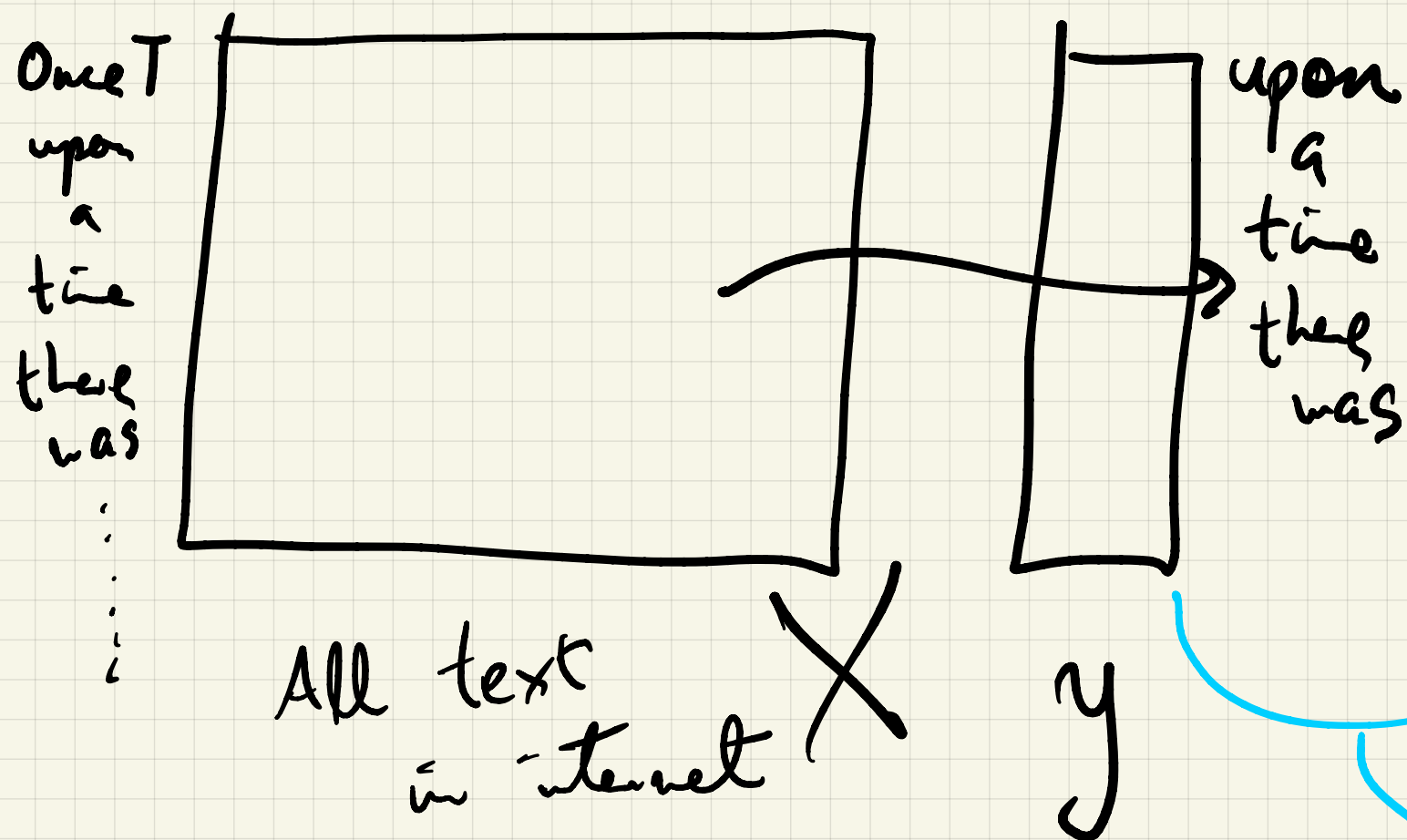
Masked self-attention



2) Model



1) Data



$\nabla \ell$

ℓ

$\hat{y}$

ℓ

How it predicts

How much I believe
that after "Once upon a..."
the word time comes next.

Transf

size of English

[EOL]

1) Data Initial embeddings

[CLS]

Once
upon
a
time

[EOL]

Guessed

upon
a
time
there
was
:

၂

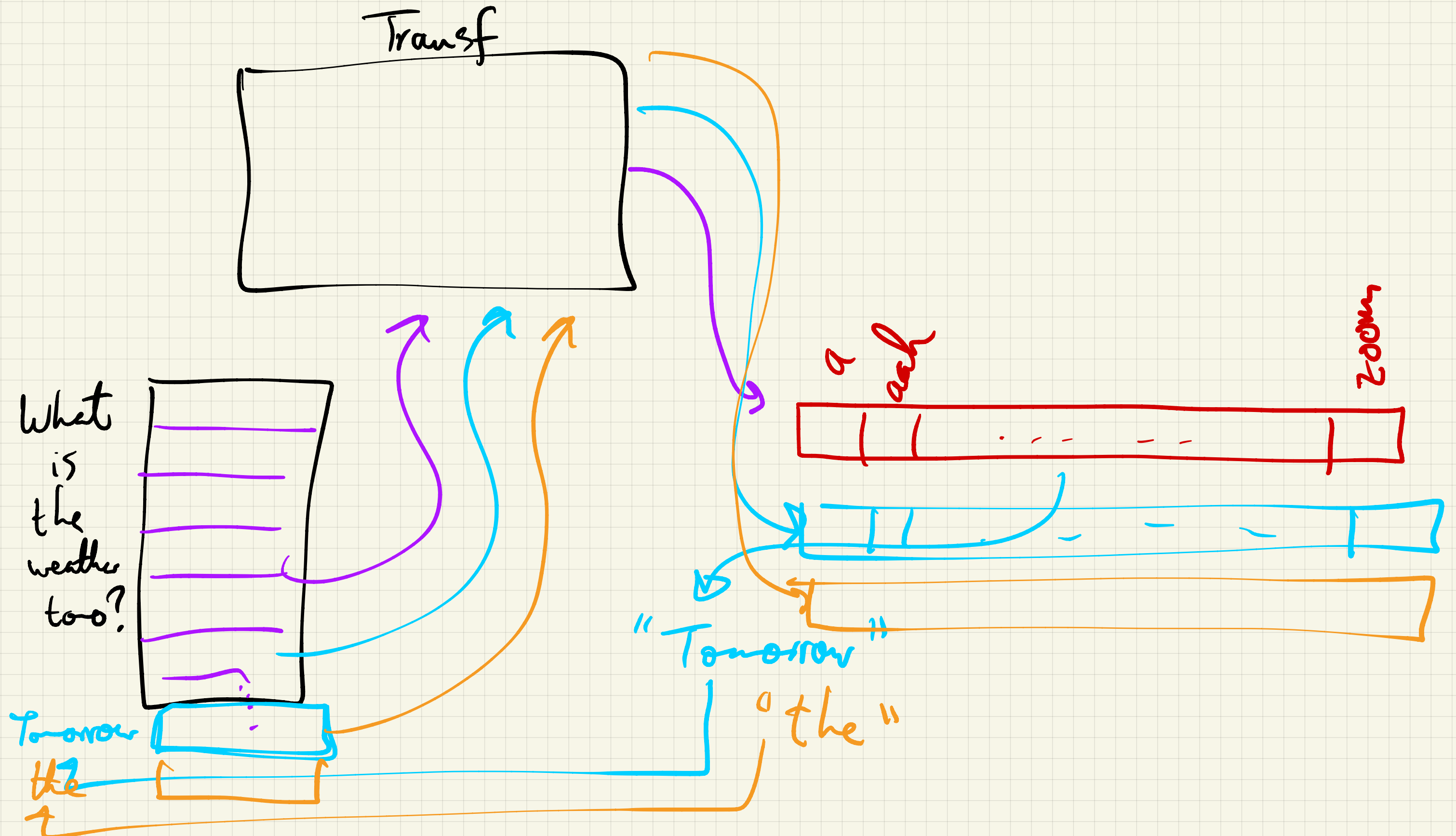
aah
abacus
...
there
line
upon
was
zoom

A hand-drawn diagram on a grid background. The grid is composed of red lines. A blue rectangle highlights a central area. Inside this blue rectangle, there are three small red marks that look like checkmarks or ticks. The blue rectangle is drawn with thick blue lines, and the red lines of the grid are visible both inside and outside the rectangle.

↑

l

How to use it after training?



Explain tokenizer

Word Tokenizer

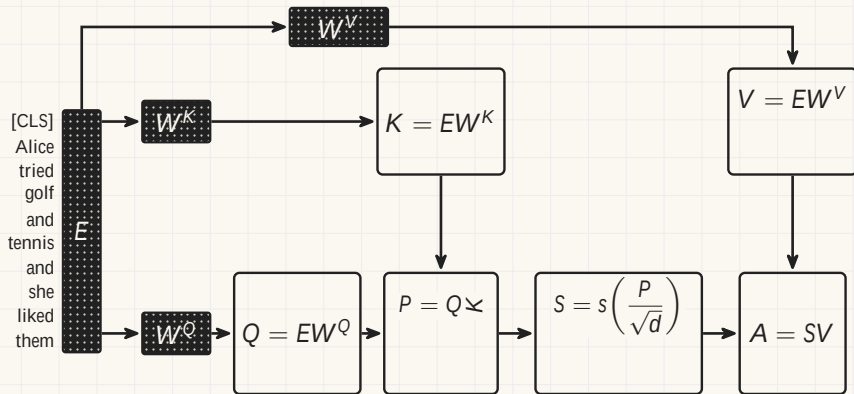
She sells seashells by the Sea

1) Define the total tokens

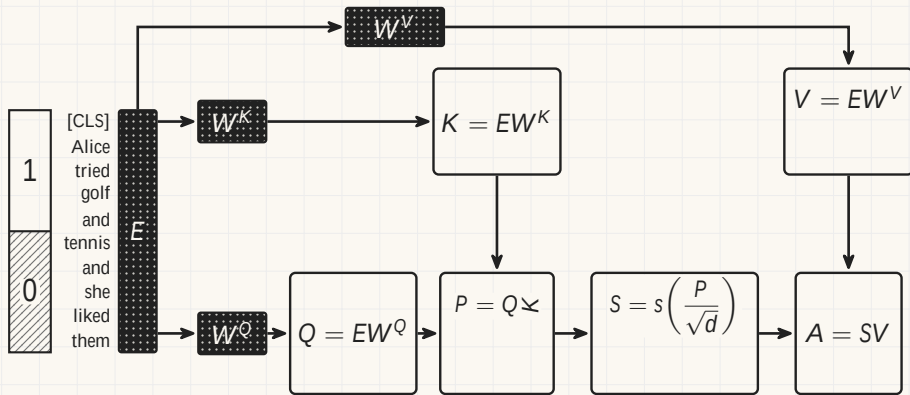
10

~~25~~
~~22~~
~~19~~
17

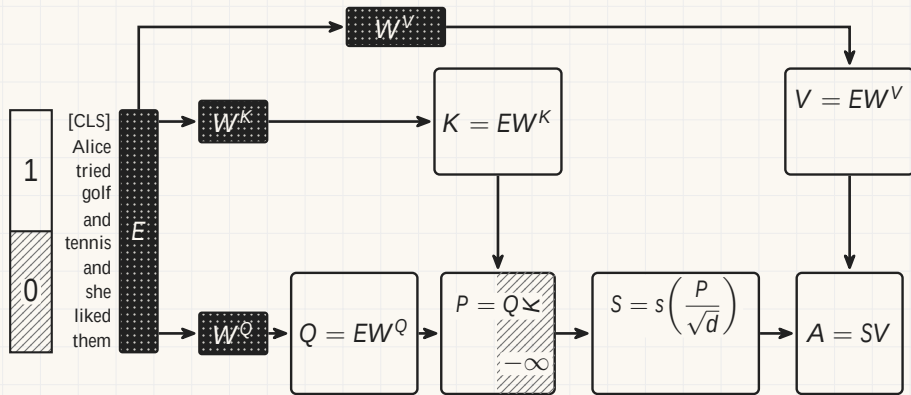
Masked self-attention



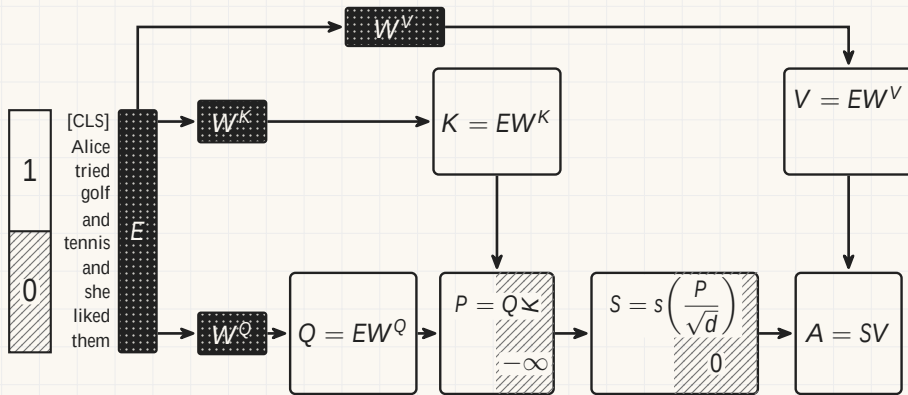
Masked self-attention



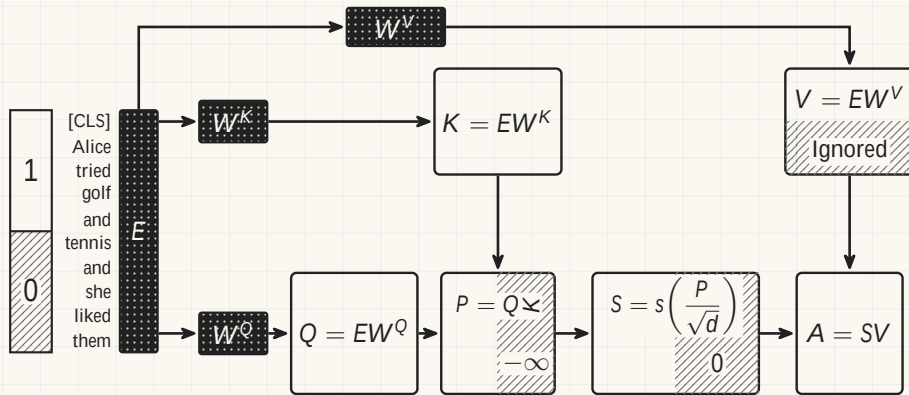
Masked self-attention



Masked self-attention



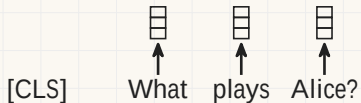
Masked self-attention



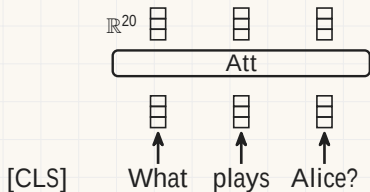
Note how the final embeddings depend only on the embeddings of the unmasked tokens!

Inside the transformer

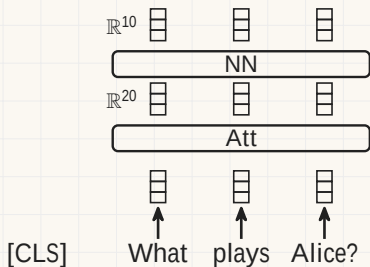
A simplified transformer architecture



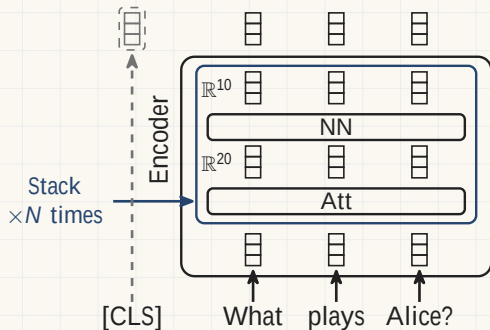
A simplified transformer architecture



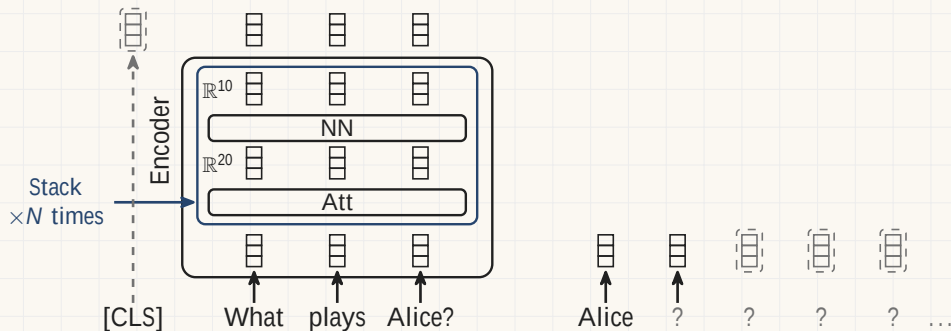
A simplified transformer architecture



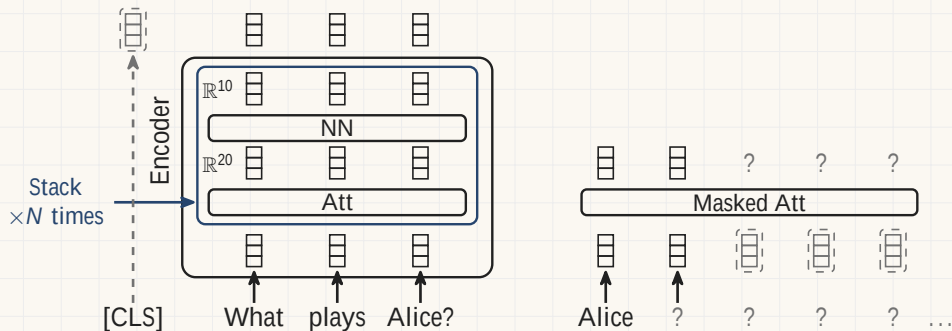
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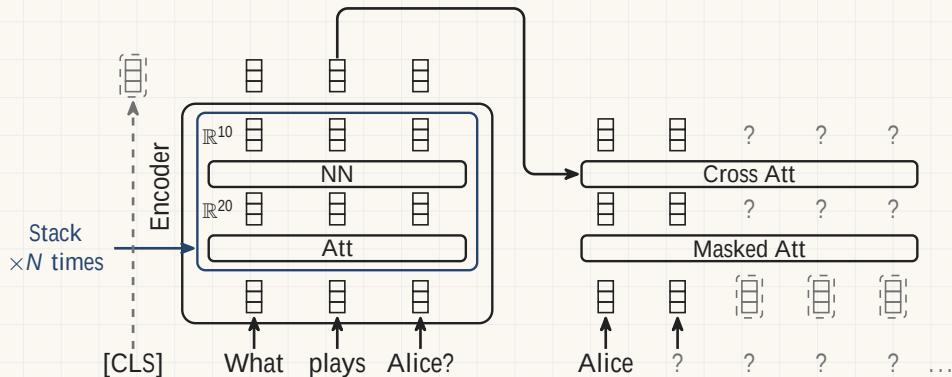
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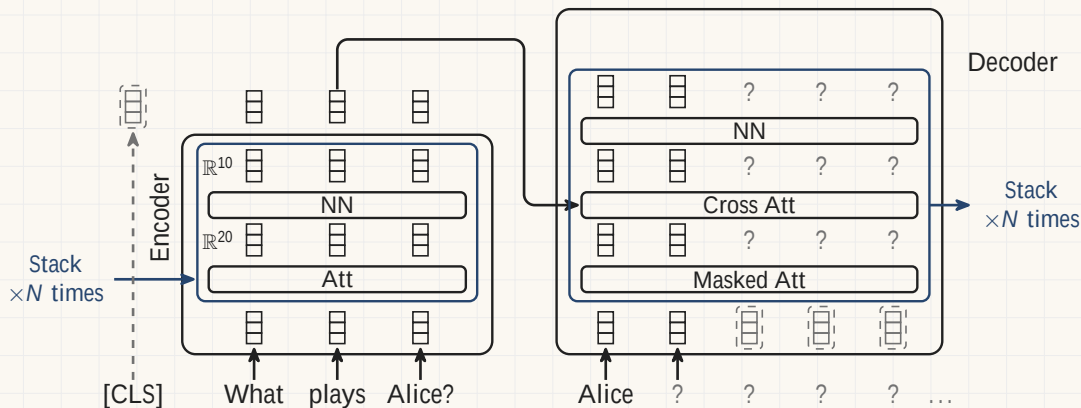
A simplified transformer architecture



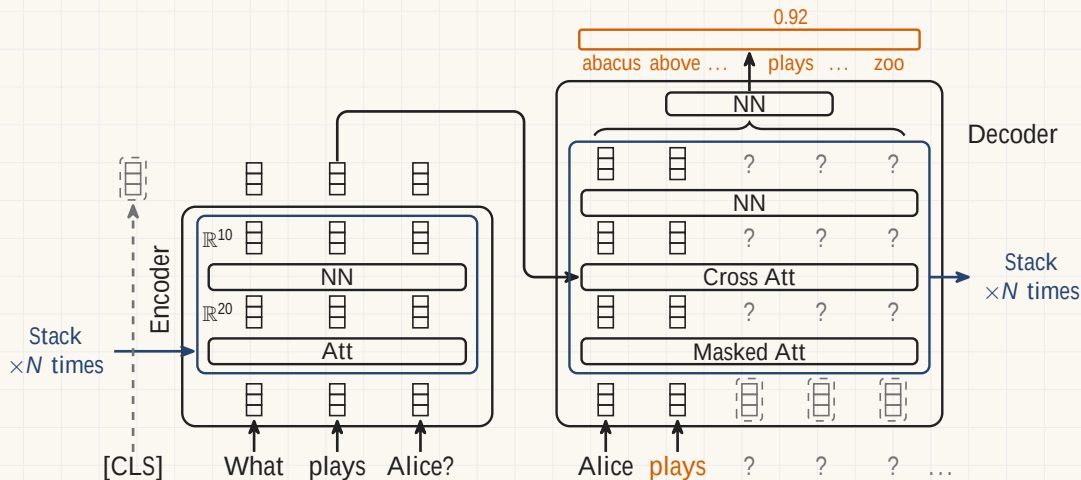
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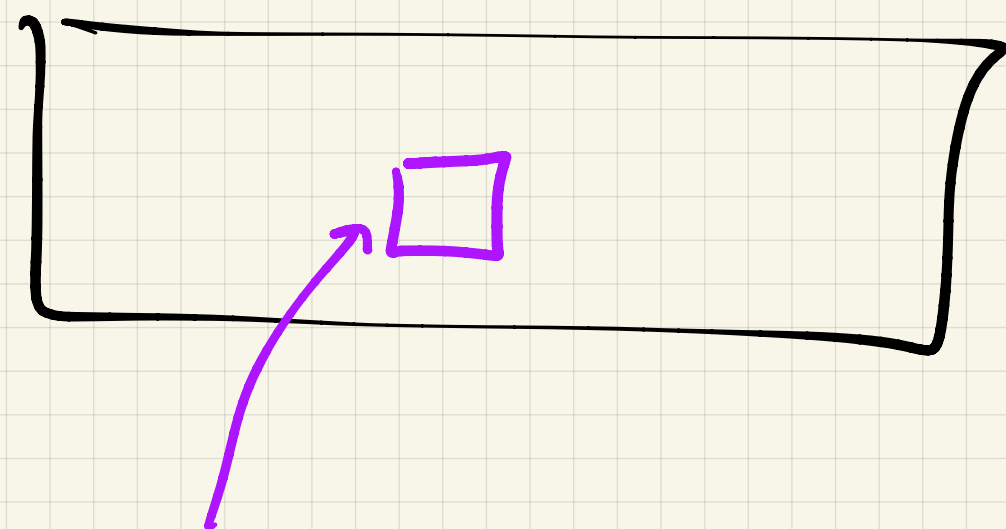


A simplified transformer architecture



Quantization

Transformer



1.456	2.378	-3.7

Quantized LLM

1.4	2.3	-3.7

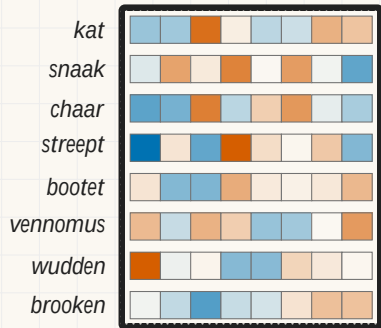
4

Section 4

Training a transformer

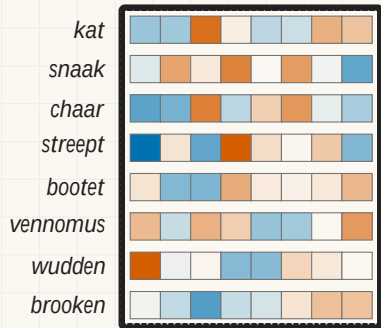
The initial embeddings

One row of numbers per word



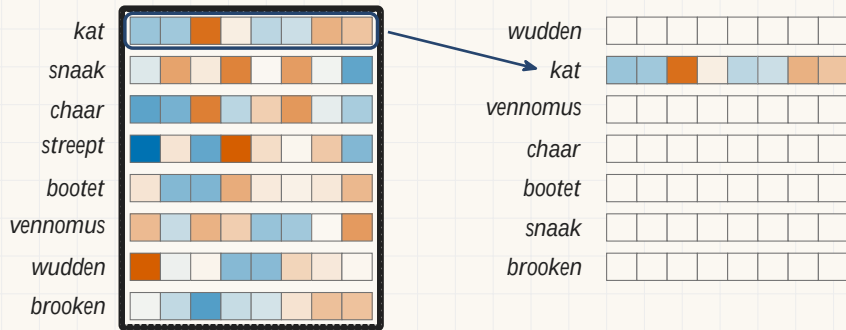
The initial embeddings

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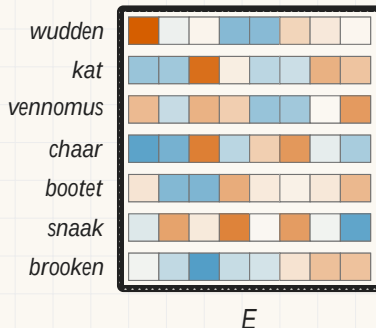
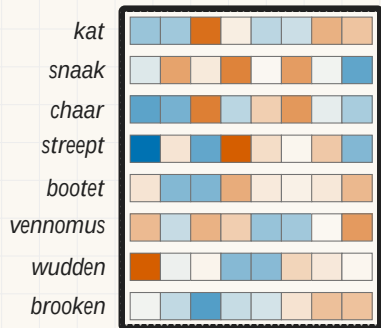
The initial embeddings

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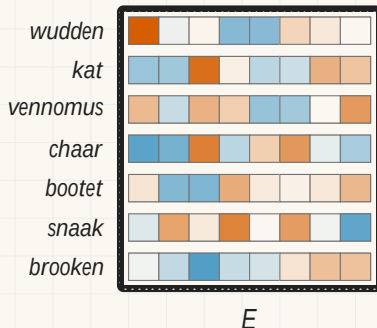
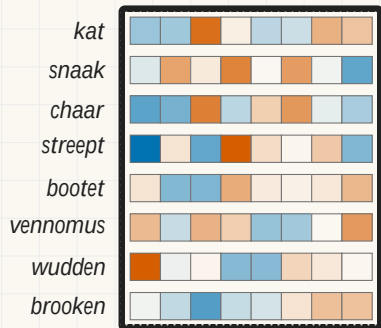
The initial embeddings

One row of numbers per word



The initial embeddings

One row of numbers per word



These numbers are weights too, like W^Q , W^K and W^V . Training guesses them

The embeddings are guessed in training

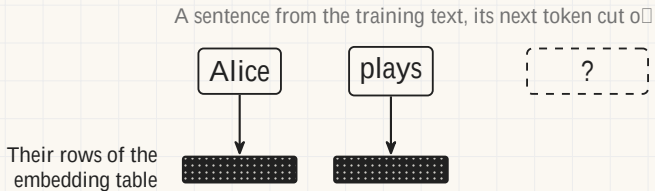
A sentence from the training text, its next token cut o□

Alice

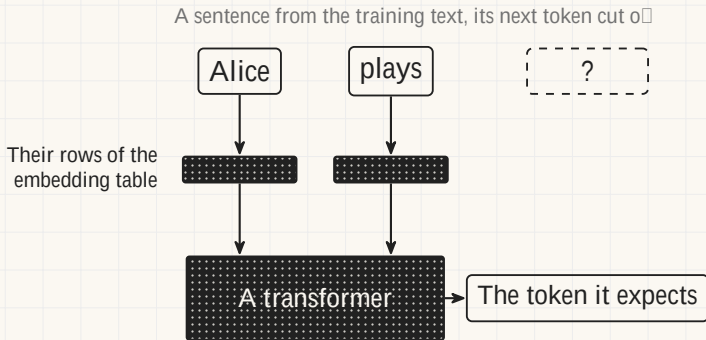
plays

?

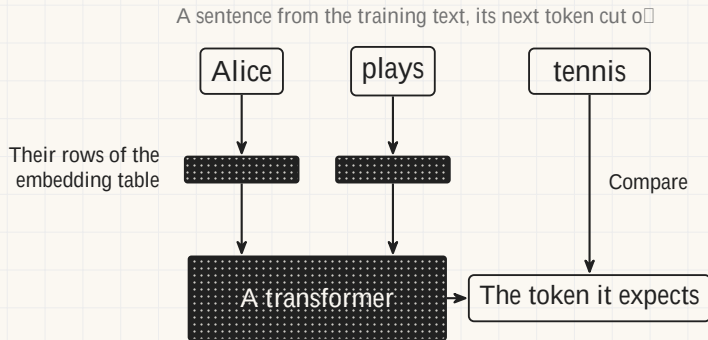
The embeddings are guessed in training



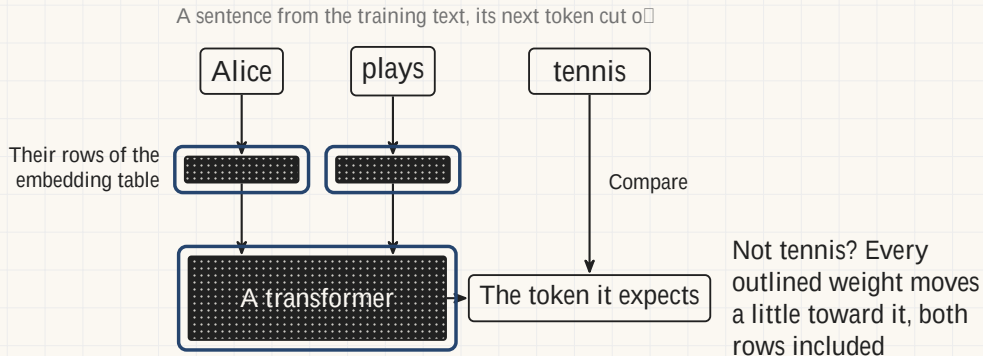
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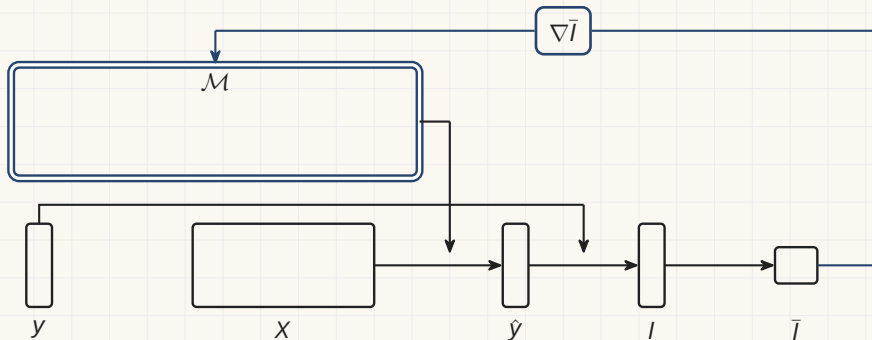
The embeddings are guessed in training



Training a transformer

A row is text with its next token cut off. Every position of every document gives one.

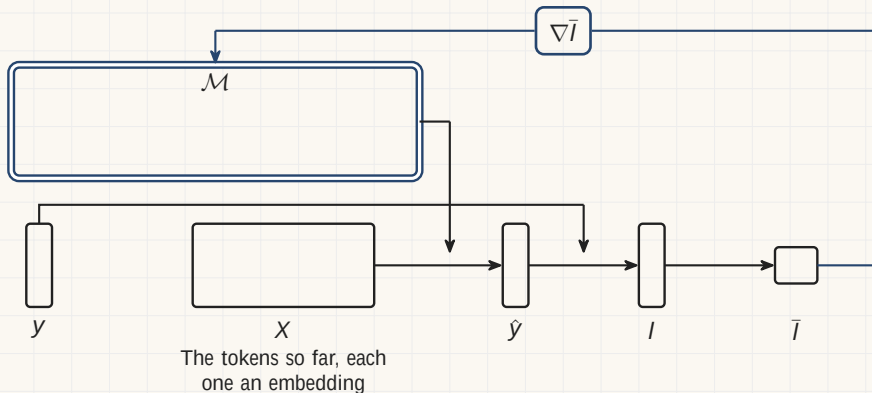
The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



Training a transformer

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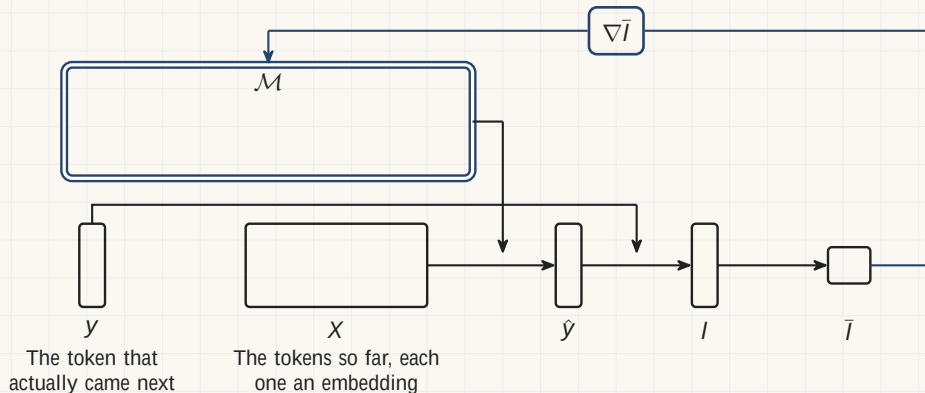
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Training a transformer

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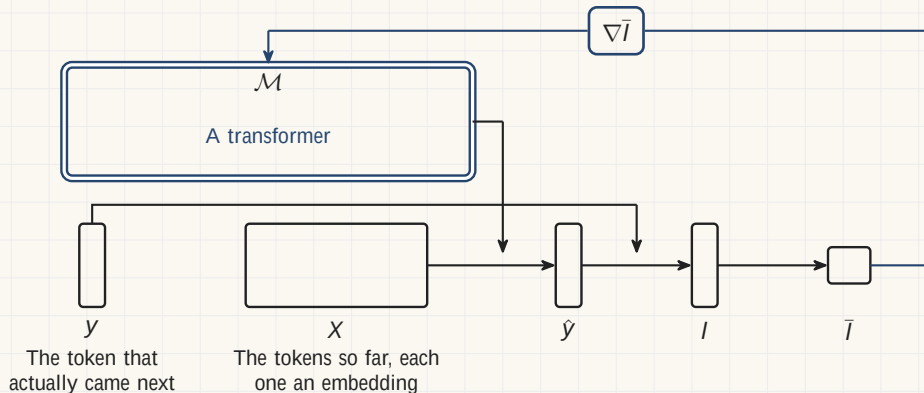
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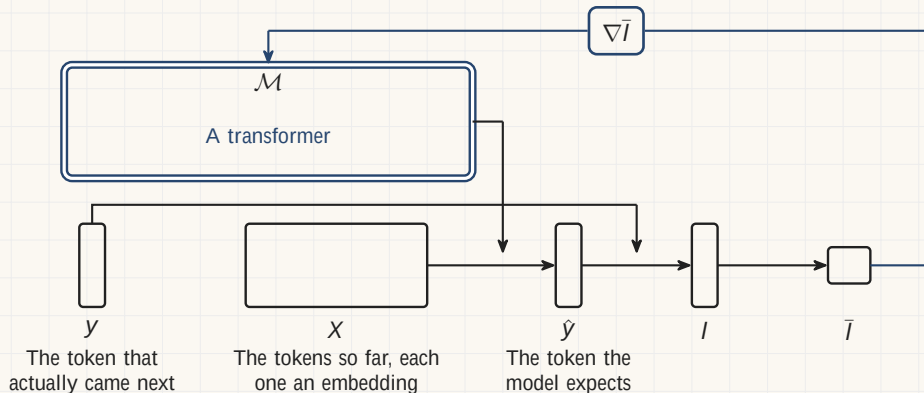
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Training a transformer

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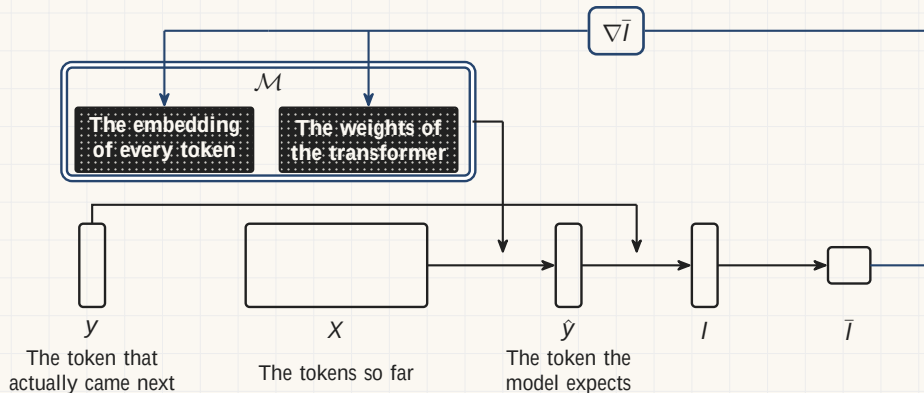
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5

Section 5

Positional encodings

Positional encodings

- Consider the sentences *the dog chased the cat* and *the cat chased the dog*.

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- They are permutations of each other, so the attention maps will also just be permutations of each other.

Positional encodings

- Consider the sentences *the dog chased the cat* and *the cat chased the dog*.
- They are permutations of each other, so the attention maps will also just be permutations of each other.
- Note that attention mechanisms do not pay attention to the position of a token in the sentence.

Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
it	0	1	0	0	0	0
was	1	1	0	0	0	0
two	0	0	1	0	0	0
tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

BP

Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
it	0	1	0	0	0	0
was	1	1	0	0	0	0
two	0	0	1	0	0	0
tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

BP

$$\begin{aligned}BP_{ij} &= (i \div 2^j) \bmod 2 \\ &= \lfloor 2^{-j} i \rfloor \bmod 2\end{aligned}$$

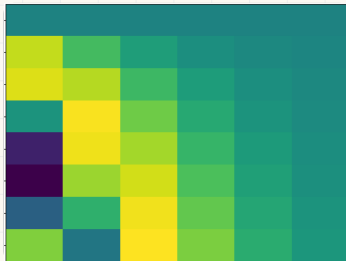
- We would like to create an embedding that encodes the position of the word in the sentence.
- A natural idea is to use a binary encoding of the position.
- Note that the entry i, j of this positional matrix has the form $f(\alpha^{-j}i)$, where $\alpha > 0$ is a scaling factor and f is a periodic function.
- In the binary positional encoding case, $f(x) = \lfloor x \rfloor \bmod 2$.

Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
it	0	1	0	0	0	0
was	1	1	0	0	0	0
two	0	0	1	0	0	0
tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

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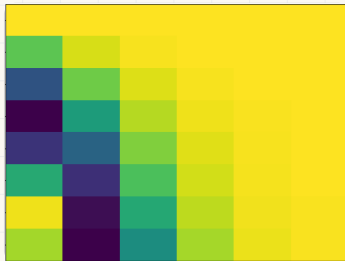
- We generalize this by using $\sin$ or $\cos$ instead of $f(x) = \lfloor x \rfloor \bmod 2$.

Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
it	0	1	0	0	0	0
was	1	1	0	0	0	0
two	0	0	1	0	0	0
tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

BP

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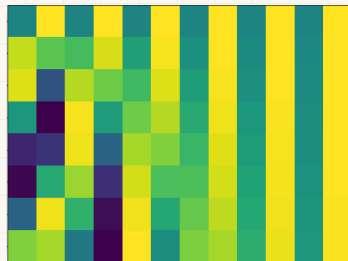
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Binary positional encoding

[CLS]	0	0	0	0	0	0
because	1	0	0	0	0	0
it	0	1	0	0	0	0
was	1	1	0	0	0	0
two	0	0	1	0	0	0
tired	1	0	1	0	0	0
[EOS]	0	1	1	0	0	0

BP

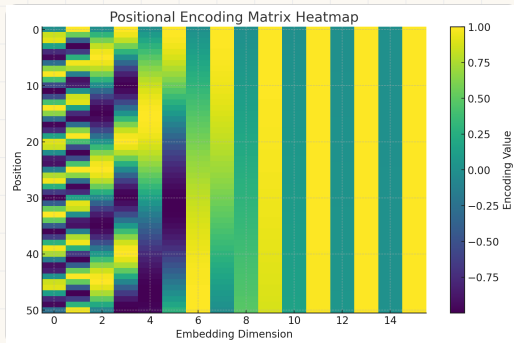
$$\begin{aligned}BP_{ij} &= (i \div 2^j) \bmod 2 \\ &= \lfloor 2^{-j} i \rfloor \bmod 2\end{aligned}$$



- We actually use both $\sin$ and $\cos$ and then interleave them.

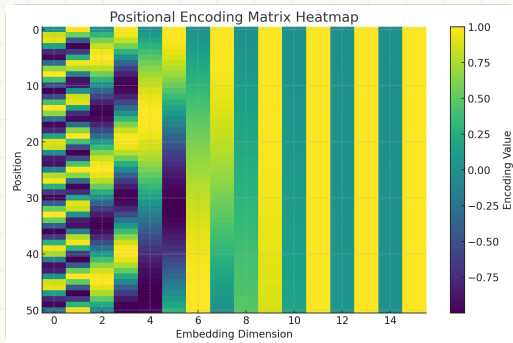
Positional encodings

- The positional encoding matrix p has T rows and d columns.



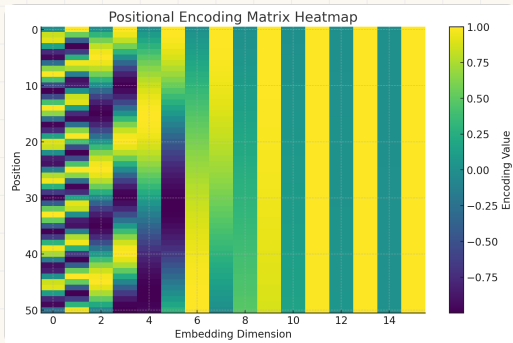
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- The value d is the dimensionality of the positional embedding.



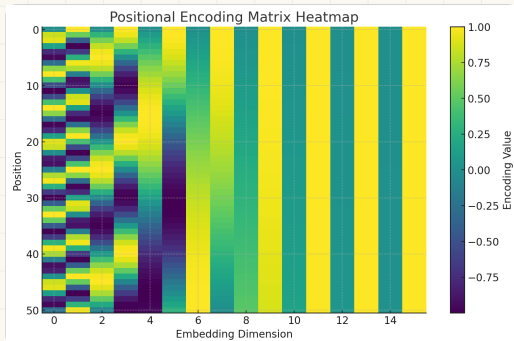
Positional encodings

- The positional encoding matrix p has T rows and d columns.
- The value T is the maximum number of tokens in a sentence.
- The value d is the dimensionality of the positional embedding.
- The encoding of token i is defined through the formulas

$$p_{i,2j} = \sin(\alpha^{-2j}i) \text{ and}$$

$$p_{i,2j+1} = \cos(\alpha^{-2j}i),$$

for $j \leq d/2$ and $\alpha = 10^{4/d}$.



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- With sufficient dimensionality, since the positional embeddings are somewhat statistically independent from the token embeddings, both embeddings are somewhat orthogonal. Hence, the addition still preserves information about the two vectors.
- Since most of the operations done in an attention mechanism are linear, the final embedding is somewhat the addition of the result of the operations applied to both the positional and the token embeddings.

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Section 6

Residual blocks

The degradation problem

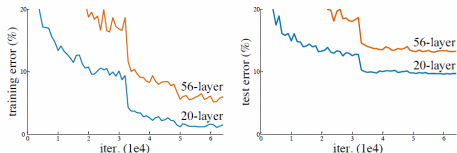


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

- When deeper networks are able to start converging, a *degradation problem* appeared. Training performance degrades rapidly. This is not because of overfitting.

Figure 1 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

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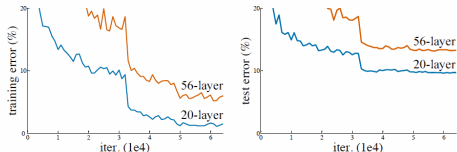


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- When deeper networks are able to start converging, a *degradation problem* appeared. Training performance degrades rapidly. This is not because of overfitting.
- The problem is mostly due to difficulties of information propagating, both forward and backward, through the layers in the network.

Residual blocks

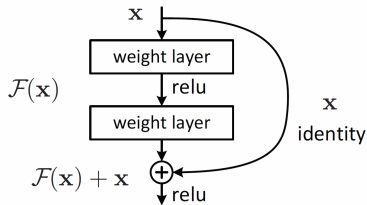


Figure 2. Residual learning: a building block.

Figure 2 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

Residual blocks

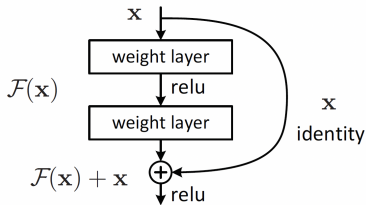


Figure 2. Residual learning: a building block.

- To address this problem, residual blocks were introduced. They work by adding the output of a layer to the output of a few layers ahead.

Figure 2 of Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun. Deep Residual Learning for Image Recognition. CVPR 2016.

<https://arxiv.org/abs/1512.03385>

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Section 7

Transformer architecture

Transformer architecture

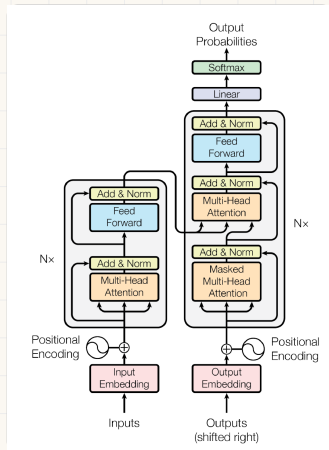


Figure 1 of

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser and Illia Polosukhin.

Attention Is All You Need.

Advances in Neural Information Processing Systems 30 (NeurIPS 2017).

<https://arxiv.org/abs/1706.03762>